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Construction and Empirical Study of Music Education Model Based Upon Big Data and Relevance Theory

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ABSTRACT

In the era of big data, music education faces the challenge that traditional teaching modes cannot meet students' individual needs. This study aims to integrate big data technology with Connectivism learning theory to construct and validate a new "big data + music education" teaching model. K-means clustering, linear regression, and Apriori association rule mining are applied to systematically evaluate the effectiveness of the model. The study develops a student-centered, data-driven, and intelligently matched teaching ecosystem; designs a whole-process instructional framework covering pre-class, in-class, and post-class stages; and establishes a multidimensional dynamic evaluation mechanism based on process data. This study verifies the practical value of big data technology in improving instructional precision and personalization in music education while providing a replicable and scalable theoretical framework and practical reference for art education reform in the context of educational informatization.

KEYWORDS

Big Data, Relevance, Music Education, Teaching Mode, Learning Analysis, Personalized Learning

INTRODUCTION

With the rapid development of information technology, the enormous value embedded in massive data has gradually been explored and utilized, providing new opportunities for educational reform (Li et al., 2025; Qu, 2021; Saif et al., 2022). Music education, as a key approach to cultivating students' aesthetic literacy and artistic talent, also faces profound impacts and new challenges in the big data era (Lan, 2022; Min, 2025; Teng, 2025). On the one hand, students' music learning needs increasingly demonstrate diversification and personalization (Fang & Luen, 2024; Lin & Chen, 2025). They are no longer satisfied with traditional standardized teaching modes but instead seek customized music education experiences aligned with their interests, talents, and learning progress (Amiri et al., 2024). On the other hand, the expanding variety and volume of music education resources have increased their dispersion and complexity (Wang, 2025; Wu, 2022; Zhao, 2022). Therefore, screening, integrating, and accurately delivering appropriate teaching resources has become an urgent issue to address.

In this context, connectivism theory offers a new direction for the development of music education. The theory emphasizes the connectivity and networking of knowledge and posits that learning is a dynamic process in which individuals expand cognition by establishing connections within complex

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knowledge networks. Guided by this perspective, students are no longer passive recipients of knowledge but instead construct their own music knowledge systems through interaction with diverse knowledge sources, learning communities, and others.

The purpose of this study is to examine how big data technology can be organically integrated with connectivism theory to develop an innovative "big data + music education" teaching model. By fully leveraging the analytical capabilities of big data, it becomes possible to accurately capture students' learning behaviors, interests, preferences, and levels of knowledge mastery, thereby providing strong support for constructing a connectivist learning environment. This approach helps optimize the music teaching process and improve instructional effectiveness while offering students a more personalized, interactive, and effective learning experience. It is of significant theoretical and practical value for advancing the innovative development of music education in the contemporary era and for cultivating music talents with innovative capacity and comprehensive literacy.

LITERATURE REVIEW

At present, artificial intelligence (AI), learning analytics, and educational big data technologies are profoundly reshaping the ecology of teaching and learning (Kamalov et al., 2023; Luan et al., 2020; Tedre et al., 2021). The traditional teacher-centered music teaching model, characterized by standardized output, has become increasingly constrained in meeting contemporary literacy demands, including individualized development, creativity cultivation, and cross-cultural understanding (Zhang & Yi, 2021; Zhang et al., 2025). In recent years, the field of music education has actively explored technology integration, such as intelligent accompaniment systems (for example, AI-based piano error correction), virtual ensemble platforms, and immersive music experiences based on virtual reality (Zhao & Cheng, 2025).

Yu et al. (2023) highlighted how AI enriched music education by enabling personalized instruction and enhancing student engagement through the integration of intelligent technologies with traditional teaching practices. Lan (2022) examined the transformative potential of internet-based big data in reshaping music education, emphasizing its role in modernizing pedagogical models, preserving traditional Chinese musical heritage, and supporting broader educational reform. Hu (2023) developed a piano teaching system enhanced by big data-driven digital audio decoding to streamline processes and enrich instruction through structured project management and integrated digital course resources. Wei and Sun (2024) introduced a bidirectional long short-term memory-based Musical Instrument Digital Interface performance evaluation system that leveraged network big data and deep learning to reduce subjective bias, enhance student autonomy, and alleviate instructional burden in piano teaching. Gong (2022) developed a data-driven vocal performance teaching platform using a deep recurrent neural network to objectively evaluate and optimize national vocal music instruction, thereby improving student engagement and learning outcomes in higher education. Liang (2025) demonstrated that digital tools and online platforms enhanced creativity in vocal ensembles by fostering collaboration, sound experimentation, and flexible learning, providing a robust framework for modernizing music pedagogy. Sun (2024) investigated a hybrid convolutional neural network–recurrent neural network model applied to virtual reality-based music instruction, showing that it captured multimodal learning dynamics and significantly improved musical proficiency, theoretical understanding, and student engagement compared with conventional approaches. Li (2025) demonstrated that a virtual–real interactive music education simulation enhanced students' understanding of instrument mechanics, expressive capabilities, and learning speed, outperforming traditional instruction and highlighting its potential as a future-oriented pedagogical approach. Wei et al. (2025) proposed a 6G-enabled, edge intelligence-driven context-aware architecture that dynamically adapted to learners' needs in ubiquitous music education environments, significantly improving task efficiency and learning accuracy through AI-optimized wireless resource management.

In exploring the use of technology to empower music education, existing research has largely focused on the application of single tools, such as intelligent scoring systems or virtual musical instruments, while lacking integrated analysis of data across the entire learning process. At the same time, most empirical studies have been constrained by small sample sizes and limited contexts and have seldom integrated learning theories, such as connectivism, with big data technology, resulting in limited scalability and theoretical depth in their teaching models.

In contrast, this study demonstrates three key advantages. First, guided by connectivism theory, it constructs a trinity teaching framework that integrates data-driven approaches, connection-oriented learning, and personalized feedback, thereby strengthening the model's theoretical foundation. Second, a mixed-methods approach, combining multidimensional behavioral data mining with qualitative interviews, enables a comprehensive and multidimensional understanding of the learning process and its outcomes. Third, the study not only identifies key behavioral factors and effective learning pathways that influence academic performance but also provides preliminary evidence of the model's practical value in enhancing student participation and satisfaction, offering empirical support for the paradigm transformation of music education in the intelligent era.

MATERIALS AND METHODS

Overview of Relevance Theory

Connectivism, as an important learning theory, was proposed by George Siemens and others (Siemens et al., 2020). It posits that knowledge does not exist in isolation within an individual's mind but is distributed across a complex network of interconnected knowledge nodes. These nodes may include individuals, groups, organizations, databases, and books, while the links connecting them represent associations of knowledge. The formation and development of knowledge depend on interactions and information flow among these nodes, emphasizing connectivity and network construction while challenging the static and isolated assumptions of traditional knowledge perspectives.

From the perspective of the learning process, connectivism conceptualizes learning as a continuous and dynamic process. In the contemporary era of information abundance, knowledge evolves rapidly, and individuals cannot rely solely on fixed knowledge stored in memory to address complex and changing real-world situations. Therefore, learners must develop the ability to navigate and actively engage with diverse knowledge sources and learning communities, continuously acquiring new information and integrating it with existing knowledge systems to support ongoing expansion and updating of understanding. For example, in music learning, learners are no longer limited to knowledge derived from textbooks or teacher instruction but can access diverse resources through online music platforms, forums, social media communities, and music production software. By connecting these fragmented sources with existing musical skills and theoretical knowledge, learners can construct a more comprehensive, in-depth, and personalized music knowledge network.

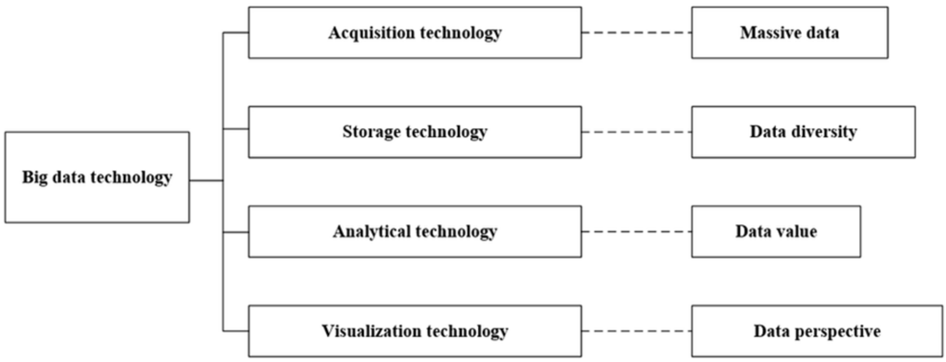
In terms of educational application, connectivism has exerted multiple influences on traditional educational models. It encourages educators to reconsider disciplinary boundaries, recognize the interdisciplinary and integrative nature of knowledge, and promote exploration of connections across disciplines, thereby enhancing students' ability to solve practical problems. At the same time, the theory emphasizes the practicality and contextual nature of knowledge, advocating for the creation of authentic and diverse learning environments in which students can apply knowledge in specific contexts and deepen their understanding.

Guided by this perspective, the role of teachers has gradually shifted from that of a knowledge transmitter to that of a guide, facilitator, and organizer of the learning process. Teachers are responsible for supporting students in building effective knowledge networks and fostering self-directed and lifelong learning abilities. Meanwhile, students become active participants in learning, engaging in the exploration and construction of knowledge, interacting and collaborating with members of the learning community, and collectively creating and sharing knowledge.

Analysis of the Potential Application of Big Data in Music Education

Big data has demonstrated significant potential in music education, characterized by its volume, variety, and value, which are reflected in the technical functions illustrated in Figure 1.

Figure 1. Big Data Technology and Its Characteristics



In terms of data volume, big data acquisition technologies use various sensors, online learning platforms, and teaching management systems to collect information on student activity records, classroom interactions, and music learning software usage (Ang et al., 2020; Liu & Yu, 2023). These data also encompass students' learning behaviors and the utilization of teaching resources, thereby generating a substantial body of music education–related data and providing a strong foundation for comprehensive and in-depth analysis (Chen, 2024). Data variety is reflected in storage technologies that accommodate multiple data types, including structured student performance data, semi-structured instructional materials, and unstructured audio and video recordings of student performances. This ensures data security and accessibility while capturing a multidimensional representation of music education.

The value of data is demonstrated through analytical techniques, such as data mining and machine learning, which identify patterns within large, diverse, and rapidly generated datasets and extract actionable insights to support evidence-based decision-making in music education. Visualization technologies further enhance this process by presenting complex data through intuitive charts and graphs. For example, bar charts can illustrate students' preferences for different music styles, while line charts can depict performance trends over time, enabling educators and students to interpret results effectively and inform teaching practice.

In the field of music education, big data is transforming multiple aspects of teaching. Its application scenarios are diverse, encompassing student learning behavior analysis, optimization of teaching resources, and evaluation and feedback processes, thereby injecting new vitality into educational development.

Analysis of Student Learning Behavior

With the support of big data, music educators can track students' learning trajectories across multiple dimensions. By analyzing indicators such as online learning duration, sequence of course module access, frequency and accuracy of practice, and participation in discussions, educators can gain precise insights into students' learning habits and interests.

Optimization of Teaching Resources

Big data enables in-depth analysis of the use of music teaching resources. By collecting data on click-through rates, downloads, dwell time, and user evaluations of textbooks, audio and video materials, and online courses, educators can identify the most effective resources and deliver them according to students' learning progress and needs.

Teaching Evaluation and Feedback

A diversified evaluation system can be established based on big data, extending beyond traditional exam-based assessment. In addition to academic performance, factors such as participation in learning activities, interaction and collaboration with peers, and creative performance in music composition or execution can be incorporated. Continuous collection and analysis of multidimensional data provide comprehensive, objective, and real-time feedback for both teachers and students. Teachers can adjust instructional strategies, address common learning challenges, and offer targeted support, while students can recognize their strengths and weaknesses, identify areas for improvement, and enhance learning motivation, ultimately improving the overall quality of music education.

Construction of the Big Data+Music Education Teaching Model Based on Connectivism Theory

Under the context of educational reform driven by the deep integration of big data and artificial intelligence, music education faces a critical opportunity to shift from standardized instruction to personalized development. Traditional teaching models cannot effectively address students' highly differentiated needs in music perception, skill acquisition, and artistic creation, while fragmented digital resources lack coherent organizational logic and connection mechanisms. Connectivism provides important theoretical guidance, emphasizing that knowledge is not statically stored in individual minds but distributed across a dynamic network of people, tools, resources, and practices. Learning is understood as the process of establishing and optimizing these connections. Based on this perspective, the "big data + music education" teaching model extends beyond the application of individual tools or algorithms and instead aims to construct an intelligent educational ecosystem centered on learners, linked by data, and driven by connectivity. This system integrates dispersed learning behaviors, teaching resources, and creative outputs, enabling knowledge to flow, evolve, and regenerate within the network, thereby supporting a shift from passive learning to active exploration and collaborative creation.

The technical implementation of this teaching model relies on a hierarchical, decoupled, and scalable platform architecture for big data–driven education. The foundational layer is a unified data access layer that connects heterogeneous teaching terminals through standardized Application Programming Interface (API) interfaces and message queues, such as Kafka or RabbitMQ. Outputs from smart piano MIDI devices, interaction logs from online platforms, and audio recordings uploaded via mobile devices are converted into event streams that conform to educational data standards, such as xAPI or Caliper, ensuring data integrity and temporal consistency. The intermediate layer functions as a data lake that integrates resources and behavioral data. Distributed file systems, such as object storage, are used to store raw data, while metadata management engines annotate the context of each learning record. For example, a solfeggio error event can be linked to a specific score identifier, tonal complexity, daily practice duration, and historical accuracy trends, thereby providing a semantically rich context for subsequent analysis.

At the level of knowledge organization, the system adopts a lightweight ontology modeling approach to construct a knowledge map in the field of music education. Unlike the complex reasoning mechanisms of general knowledge graphs, this ontology design emphasizes practicality and operability. Core entities include learners, teaching resources, music skills, creative works, and teaching activities, while relationships are defined using verbs aligned with teaching practice, such as practicing, referencing, and mastering. Resource labels are not predefined solely by experts but are collaboratively generated through teacher input, student tagging, and system recommendations. For example, when multiple students independently assign keywords such as lyricism, triplet difficulty, and pedal control after completing the first movement of Beethoven's Moonlight Sonata, the system increases the weight of these labels and associates them with the corresponding music resource node, forming a dynamic semantic enhancement mechanism based on collective input. This co-constructed and shared tagging ecosystem represents a technical realization of the connectivist view that knowledge is distributed across networks.

As the hub connecting learners with knowledge networks, the personalized recommendation engine relies on context-aware association path matching. Rather than directly predicting user preferences, the system calculates, in real time, the similarity between a learner's current state and historically successful learning pathways. For instance, when a student studies jazz harmony, the system retrieves records of learners who successfully completed the module, extracts common behavioral sequences, and dynamically recommends the most appropriate sub-path. This process also incorporates contextual variables, such as the student's available equipment (for example, access to a MIDI keyboard), available time, and willingness to collaborate. Personalized recommendation thus moves beyond one-way delivery, offering multiple connection pathways that encourage learners to actively explore and construct their own knowledge networks.

Crucially, the entire system incorporates an ecological evolution mechanism driven by bidirectional feedback. On the one hand, teachers can monitor knowledge connection density, resource circulation patterns, and areas of weak connectivity through visual dashboards, enabling them to adjust instructional focus or supplement guiding resources accordingly. On the other hand, students' new behaviors, such as bypassing recommended paths or independently exploring less common composers, are recorded as negative samples or innovative pathways to refine the confidence thresholds of subsequent association rules. In addition, the system supports the automatic integration of high-quality student-created works into the resource pool, with creator-authorized open labeling allowing these works to become new nodes within other learners' networks. This process not only facilitates the reuse of learning outcomes but also enables the system to achieve self-development and maintain diversity, similar to a biological ecosystem.

Ultimately, this technical architecture represents not a closed intelligent teaching product but an open, collaborative, and continuously evolving digital ecosystem for music education. It preserves teachers' instructional autonomy while providing students with the freedom to connect and create. At the same time, it consolidates dispersed individual practices into a shared and iterative network of collective intelligence through big data infrastructure. Guided by connectivism, technology functions not merely as an external tool but as an embedded medium of connection and a driving force for knowledge evolution, thereby establishing a new paradigm of intelligent music education that integrates teaching, learning, creation, and evaluation.

Data and Analysis Methods

K-means clustering is an unsupervised machine learning method used to automatically partition a set of objects into multiple clusters based on feature similarity. Its core principle is to maximize similarity among objects within the same cluster while maximizing differences between objects in different clusters. In this study, K-means is applied to classify students' learning data to identify distinct learning styles and needs, thereby enabling the adjustment of teaching strategies (see Equation 1).

$$J = \sum_{i=1}^{k}\sum_{j=1}^{n}\left\| x_j^{(i)} - u_i \right\|^2 \tag{1}$$

Where, J is the objective function, k is the number of clusters, $x_j^{(i)}$ is the j-th sample in the i-th cluster, and u_i is the center of the i-th cluster.

Regression analysis is a statistical method used to examine the relationship between one or more independent variables, such as learning time and platform usage frequency, and a dependent variable, such as final grades. It helps determine whether specific factors influence outcomes, as well as the magnitude and direction of these effects. This method is commonly applied to predict or explain the strength and direction, whether positive or negative, of relationships between variables. In this study, linear regression is used to analyze factors affecting students' academic performance, including study time and level of participation (see Equation 2).

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \varepsilon \quad (2)$$

Where, Y is the dependent variable (such as student grades), $X_1, X_2, \cdots, X_n$ is the independent variable (such as learning time, participation, etc.), β is the regression coefficient, and ε is the error term.

The Apriori algorithm is a data mining method used to identify frequent itemsets and association rules. It is primarily applied to discover combinations of items that frequently occur together within large datasets. For example, in retail data, it may reveal that customers who purchase beer often also buy potato chips. In educational contexts, it can identify patterns such as students who view specific instructional videos frequently downloading related exercises, thereby uncovering underlying relationships between learning behaviors. In this study, the Apriori algorithm is used to mine association rules among students' learning behaviors and to determine which activities most effectively support learning outcomes (see Equation 3).

$$\text{Support}\left(A\right) = \frac{\text{Count}(A)}{N}, \text{Confidence}\left(A \rightarrow B\right) = \frac{\text{Count}(A \cap B)}{\text{Count}(A)} \quad (3)$$

Where N is the total number of transactions, and Count (A) is the number of transactions containing itemset A.

A control group using traditional teaching and an experimental group using big data–based instruction were established to compare students' learning outcomes. Evaluation was conducted based on academic performance, classroom participation, and activity on the online learning platform. Academic performance was measured quantitatively through final examination scores and homework results. After the experiment, a satisfaction survey was administered to assess both students' and teachers' acceptance of and satisfaction with the teaching model.

In addition, a questionnaire was designed to collect students' feedback on music education and learning outcomes. Before the formal study, a pilot test was conducted to ensure the reliability, validity, and applicability of the instrument. Thirty sophomore music majors who did not participate in the main experiment were selected as pilot participants, and their demographic characteristics were consistent with those of the formal sample. After completing a simulated teaching session, participants were asked to complete the initial questionnaire and provide brief open-ended feedback regarding item clarity, response difficulty, and overall process. At the same time, brief interviews were conducted with selected participants to determine whether their understanding of key items, such as "the recommended content of the platform meets my learning needs," aligned with the intended meaning.

Based on the feedback, two ambiguous terms were revised, for example, replacing "system push" with "learning content recommended by the platform," and one item with low expected factor loading and high participant confusion was removed. Finally, Cronbach's α coefficient was calculated using the pilot data, with all dimensions exceeding 0.75, thereby preliminarily confirming the internal consistency of the questionnaire and providing methodological support for the formal study. At the same time, conduct in-depth interviews with teachers and students to understand their learning experience in the big data environment. Students and teachers of different grades and backgrounds were selected for semi-structured interviews. Discuss their learning experiences, challenges and suggestions for teaching mode in big data environment.

RESULT AND ANALYSIS

To ensure baseline comparability between the experimental and control groups, this study recorded and matched students' demographic characteristics and music literacy levels prior to the intervention. No significant differences were observed between the experimental group ($n = 70$) and the control group ($n = 68$) in grade composition, both consisting of sophomore students from the same university, gender distribution, 69.6% in the experimental group and 70.0% in the control group, or pretest scores, with means of 71.4 ± 5.2 and 70.1 ± 6.1, respectively (independent samples t test and chi-square test, $p = 0.87$). In addition, the proportion of students with prior musical instrument experience was comparable between the two groups, approximately 58% versus 60%. These results indicate that the grouping effectively controlled potential confounding variables and provided a reliable basis for subsequent comparisons of teaching outcomes.

During the intervention, students in the experimental and control groups were exposed to distinctly different teaching approaches. The experimental group adopted a data-driven teaching model supported by an intelligent learning platform that collected multidimensional behavioral data in real time, including daily learning duration, weekly login frequency, video completion rates, and forum interaction levels. Based on these data, the system employed a personalized recommendation engine that integrated contextual factors such as current learning status, device type, available time, and willingness to collaborate, dynamically delivering adaptive learning paths and resources, including targeted instructional videos, interactive exercises, and collaborative tasks. The platform also utilized a domain-specific music education knowledge map to organize resources and supported student participation in community-based activities such as peer evaluation and creative discussions. Teachers monitored the learning network through visual dashboards and adjusted instructional strategies accordingly. In contrast, the control group followed a traditional teacher-centered instructional model characterized by standardized content delivery, reliance on unified teaching materials, and one-way knowledge transmission.

Cluster Analysis of Students' Behavior

Based on real log data collected from the intelligent teaching platform, this study extracted multidimensional behavioral indicators from students' complete learning trajectories. These indicators included key variables such as average daily study time, average weekly platform logins, video completion rate, and forum posting frequency. The selection of these indicators aligns with the core elements of connection formation, node activation, and network participation in connectivism. K-means clustering was applied to classify student groups in an unsupervised manner. The optimal number of clusters was determined to be three using the elbow method and silhouette score. The clustering results were further validated through significance testing, such as analysis of variance, to ensure statistically significant differences in behavioral characteristics among the groups. The clustering results are presented in Table 1.

Table 1. Clustering Results of Students' Learning Behavior

Cluster ID	Sample Size	Average daily study time (minutes)	Average weekly platform logins	Video completion rate (%)	Forum posting frequency (posts/week)	Behavioral profile label
1	58	118	6.2	89%	3.1	Proactive explorer
2	47	36	2.0	42%	0.4	Passive complier
3	33	90	4.5	75%	1.8	Steady follower

The clustering results indicate that the three student groups exhibit clear stratification across average daily study time, weekly platform logins, video completion rate, and forum posting frequency. Cluster 1, the proactive explorer group, demonstrates the highest level of engagement, with an average daily study time of 118 minutes, 6.2 weekly logins, a video completion rate of 89%, and a forum posting frequency of 3.1 posts per week. These patterns reflect high autonomy, persistence, and deep participation. Such students not only absorb structured knowledge actively but also expand and reinforce cognitive connections through frequent community interaction, aligning with the connectivist view that learning involves establishing, activating, and maintaining effective network nodes.

In contrast, Cluster 2, the passive complier group, exhibits weak connectivity characteristics. These students average only 36 minutes of daily study time, 2 weekly logins, a video completion rate of 42%, and minimal forum participation at 0.4 posts per week. This low-frequency, fragmented, and superficial engagement hinders the formation of stable knowledge networks, resulting in a disconnect between technology access and meaningful learning. In online music education environments that rely on self-directed exploration and social interaction, such learners are unlikely to achieve satisfactory outcomes without targeted intervention.

Cluster 3, the steady follower group, represents an intermediate pattern. With an average daily study time of 90 minutes and a video completion rate of 75%, these students demonstrate basic learning discipline. However, their relatively low forum activity, at 1.8 posts per week, and moderate login frequency, 4.5 times per week, indicate limited initiative in extending learning beyond required tasks. While they maintain continuity in knowledge acquisition, they show limited motivation for deeper engagement or exploration.

Importantly, the four behavioral indicators selected in this study, average daily study time, weekly platform logins, video completion rate, and forum posting frequency, correspond directly to critical dimensions of music learning. Daily study time reflects sustained effort required for skill development, while login frequency and video completion rate indicate learning continuity and depth of knowledge acquisition. Forum participation captures essential social practices in music education, including peer evaluation, collaborative creation, and problem-solving. This multidimensional tracking approach transforms the learning process from an opaque system into a measurable and observable pathway that supports targeted intervention.

Based on these cluster profiles, differentiated instructional strategies are necessary. Proactive explorers can be provided with advanced challenges and leadership roles within learning communities. Passive compliers may benefit from early warning systems and structured, low-threshold interactive guidance to increase engagement. Steady followers can be supported through incentive mechanisms designed to encourage deeper participation and progression beyond baseline requirements.

Analysis of Influencing Factors of Academic Achievement

To further quantify the influence of students' multidimensional behavioral characteristics on academic performance in a digital learning environment, this study constructs a multiple linear regression model based on behavioral data collected from platform logs and the teaching management system. The model uses students' final grades as the dependent variable, aiming to reveal how observable learning behaviors translate into academic outcomes.

Regarding the selection of independent variables, total study time reflects individual investment in music skill development and knowledge internalization; classroom participation, including online attendance, real-time question responses, and completion of group tasks, represents engagement in structured instructional settings; and platform interaction frequency, encompassing forum posting, peer evaluation, and teacher feedback exchanges, captures students' active connectivity within the learning network. All continuous variables were standardized to eliminate scale effects, and variance inflation factor testing confirmed the absence of significant multicollinearity (VIF < 2.5). The regression model was estimated using ordinary least squares, and standardized regression coefficients (β) were reported to compare the relative contributions of each behavioral factor. The results of the analysis are presented in Table 2.

Table 2. Analysis of Influencing Factors of Academic Achievement

Independent variable (X)	Regression coefficient (β)	Standard error	t-value	p-value	Significance
Total study time (hours)	0.38	0.11	3.45	0.001	**
Classroom participation	0.52	0.09	5.78	<0.001	***
Platform interaction Frequency	0.21	0.13	1.62	0.108	
constant	58.7	4.2	13.98	<0.001	***
R^2	0.67				

Note. Significance levels: *** p < 0.001, ** p < 0.01.

The linear regression results presented in Table 2 reveal the quantitative relationship between learning behavior and academic achievement and provide insight into the differentiated contributions of various interaction behaviors. The model demonstrates strong explanatory power ($R^2 = 0.67$, $p < 0.001$), indicating that the three behavioral indicators jointly explain nearly 70% of the variation in final grades and supporting the feasibility of linking learning process data with academic outcomes.

Among the predictors, classroom participation shows the highest standardized regression coefficient ($\beta = 0.52$, $p < 0.001$), significantly exceeding the other variables. This finding has important implications for educational practice. In music education, the classroom is not merely a space for one-way instruction but a social-cognitive environment characterized by interactive activities such as improvisation, real-time peer evaluation, group collaboration, and immediate teacher feedback. Through these processes, students engage in multimodal knowledge construction via physical participation, such as performance, auditory interaction, such as ensemble work, and emotional engagement, such as sharing creative outputs, reflecting the connectivist principle that learning emerges through dynamic interactions within a network.

In comparison, although total study time also positively predicts academic performance ($\beta = 0.38$, $p = 0.001$), its effect is weaker than that of classroom participation. This suggests that while time investment is important, extended practice or prolonged video viewing without structured guidance and meaningful interaction may lead to inefficient repetition, limiting the development of transferable skills and deeper aesthetic understanding.

It is noteworthy that although platform interaction frequency, such as forum posting and commenting, shows a positive relationship with academic performance ($\beta = 0.21$), it does not reach statistical significance ($p = 0.108$). This result appears to contradict the connectivist emphasis on networked interaction but instead highlights that the quality of interaction is more important than its quantity. In the current platform design, many online interactions tend to be superficial, such as routine check-ins or template-based evaluations, lacking deeper cognitive engagement or emotional involvement and thus failing to activate meaningful knowledge connections.

Music learning, in particular, relies on embodied and contextual experience. Precise rhythm imitation and real-time harmonic adjustment can facilitate neuromuscular memory and the development of auditory schemas more effectively than text-based interaction. Therefore, the value of social interaction should not be dismissed. Rather, digital music education should prioritize the development of high-fidelity interactive environments, such as virtual ensembles, AI-assisted real-time feedback, and cross-institutional collaborative creation, instead of focusing solely on increasing the volume of digital interaction traces.

Analysis of Association Rules of Learning Behavior

To further understand the underlying logic and sequential patterns of learning pathways, the learning process was decomposed into a time series of atomized behaviors, including watching instructional videos, completing homework exercises, participating in group discussions, submitting original musical works, and viewing presentation slides. On this basis, the Apriori association rule mining algorithm was applied to systematically identify high-frequency co-occurring behavior patterns from extensive learning trajectories. By setting minimum support (min_support = 0.15) and minimum confidence (min_confidence = 0.70), the algorithm effectively filtered out incidental associations and focused on stable and generalizable behavioral patterns.

In addition, lift was introduced as a key evaluation metric (lift > 1.5) to eliminate spurious associations caused by the high occurrence of individual behaviors, ensuring that the identified rules possess statistical significance and predictive value. Table 3 presents three representative association rules characterized by high confidence and lift, with clear educational relevance. These rules illustrate effective learning pathways from foundational engagement to advanced creative activities and provide empirical support for learning path recommendation and adaptive navigation within intelligent teaching system

Table 3. Mining Results of Learning Behavior Association Rules

Rule	Antecedent → Consequent	Support	Confidence	Promotion
1	Watched instructional videos, Completed homework exercises → Final score ≥ 85	0.32	0.86	2.15
2	Participated in group discussions → Submitted original musical work	0.28	0.79	1.92
3	Only viewed PPT slides → Final score < 70	0.25	0.73	1.80

The three high-confidence association rules presented in Table 3 reveal typical behavior–outcome relationships in the digital music learning environment and illustrate the formation of effective learning pathways as well as early warning signals of ineffective patterns. Rule 1, watched instructional videos and completed homework exercises leading to a final score ≥ 85, shows a confidence of 0.86 and a lift of 2.15. This finding indicates that the integration of audio-visual input and skill-based output forms a core mechanism for achieving high-level learning outcomes. In music education, instructional videos provide multimodal information, including performance demonstrations, rhythm analysis, and expressive interpretation, while homework exercises enable skill reinforcement and immediate feedback. Together, they establish a cognitive loop of perception, connection, practice, and reinforcement, consistent with connectivist principles. Through observation, students activate cognitive nodes, and through practice, they reconstruct knowledge networks, thereby enhancing learning effectiveness and supporting the design of integrated demonstration and practice resources.

Rule 2 highlights the catalytic role of collaborative interaction in fostering musical creativity from a social perspective. Music creation, which relies heavily on subjective expression and cultural context, can be constrained by individual experience. Group discussions provide diverse perspectives, immediate feedback, and emotional engagement, which stimulate new associations and creative breakthroughs. For instance, exposure to peers' interpretations may inspire students to modify harmonic structures, reflecting knowledge reconstruction through social interaction. This process aligns with the connectivist view that learning emerges within networks rather than within isolated individuals. The rule supports the inclusion of collaborative creative modules in the platform and informs targeted interventions that guide students from imitation to original creation.

Rule 3, which links exclusive reliance on PPT materials to a final score < 70, with a confidence of 0.73 and a lift of 1.80, represents a typical ineffective learning pattern. Students who depend solely on static materials tend to achieve lower performance outcomes. Music, as a temporal and embodied art form, requires engagement with dynamic elements such as pitch, rhythm, dynamics, and timbre, which cannot be fully conveyed through text or static visuals. The absence of auditory input and experiential feedback prevents the formation of essential sensory and motor connections, resulting in knowledge remaining at an abstract level and limiting its translation into practical performance or appreciation skills.

Analysis of Learning Satisfaction

To assess students' satisfaction with the "big data + music education" teaching model, a survey questionnaire was administered to 138 students who participated in the program at the end of the semester, yielding 129 valid responses. The questionnaire was structured around four dimensions: course content, teaching methods, learning resources, and learning community engagement. Each dimension included multiple items evaluated using a five-point Likert scale, where 1 indicated very dissatisfied and 5 indicated very satisfied. The results of the survey are presented in Table 4.

Table 4. Satisfaction with the New Teaching Mode

Investigation dimensions	Very dissatisfied proportion (%)	Dissatisfied proportion (%)	General proportion (%)	Satisfied proportion (%)	Very satisfied proportion (%)	Average Score (%)
Content of courses	0	2.8	16.7	44.4	36.1	4.1
Teaching method	0	4.4	13.9	41.7	40.0	4.0
Learning Resource	0	1.7	11.1	47.2	40.0	4.2
Learning community engagement	1.1	2.8	16.7	38.9	40.5	4.1

Table 4 indicates that students' satisfaction demonstrates an overall positive trend, with average scores across the four dimensions ranging from 4.0 to 4.2 out of 5. The proportion of satisfied and very satisfied responses is consistently high, suggesting that the "big data + music education" teaching model grounded in connectivism has been widely recognized at the experiential level. Among the dimensions, learning resources received the highest average score, 4.2, indicating strong student approval of the platform's multimodal resources, including demonstration videos, interactive exercises, and expanded music libraries. This positive evaluation is supported by qualitative interview data. Many students reported that demonstration videos enabled repeated observation of fingering and breathing patterns, offering greater clarity than in-class demonstrations. Others noted that the system provided targeted training based on practice errors, functioning similarly to a personalized teaching assistant. These responses indicate that the model effectively addresses key requirements in music learning, including repeatability, visualization, and immediate feedback, through multimodal resources and personalized learning pathways.

However, some students with weaker foundational skills reported difficulty navigating the abundance of resources. As noted in interviews, the availability of extensive materials sometimes led to uncertainty about where to begin, particularly when prerequisite knowledge had not been fully mastered. This highlights a tension between resource richness and insufficient guidance in learning pathways. Although big data enables the provision of extensive content, the absence of structured navigation or staged learning goals may increase cognitive load, particularly for students with lower self-regulation. This finding is consistent with the behavioral clustering analysis, in which passive complier students exhibited limited ability to effectively utilize available resources despite operating within the same platform environment.

In addition, the dimensions of teaching methods, with an average score of 4.0, and course content, with an average score of 4.1, were also positively evaluated. These results indicate that students recognize the effectiveness of this teaching model, which moves beyond the limitations of traditional teacher-centered, one-way instruction by incorporating data-driven personalized tasks, tiered challenges, and integrated online and offline learning designs. In music education, the flexibility of this approach allows students with different skill levels to adopt individualized learning rhythms and expressive styles, thereby enhancing their sense of autonomy and control. In the dimension of learning community engagement, with an average score of 4.1, the quantitative data appear favorable, but interview findings reveal a more complex picture. Although nearly 80% of students reported satisfaction, some noted that their posts often received no response, leading to reduced motivation for interaction over time. Group discussions were sometimes perceived as routine tasks, with minimal meaningful participation. This observation helps explain why platform interaction frequency did not significantly predict academic performance in Table 2. Nevertheless, students who actively engaged in community activities reported substantial benefits. For example, one student described collaborating on the creation of an electronic folk song, with different members responsible for arrangement, orchestration, and performance, an experience that fostered a transition from learner to creator.

Implementation Challenge Analysis

Students' Technical Adaptation Obstacles

In the early stage of teaching implementation, many students showed clear adaptation difficulties when using the intelligent learning platform. In particular, some students with weak digital skills felt at a loss when faced with various functional modules on the platform (such as personalized resource recommendations, knowledge map navigation, work uploading, and the mutual evaluation system) and even experienced frustration due to complex operations, which reduced their subsequent participation. To address this problem, the research team organized a dedicated operation training workshop before the formal start of the course and guided students to become familiar with the basic platform processes through simulated tasks. At the same time, an illustrated user manual and a series of micro-video tutorials were developed and embedded in the platform to provide instant help tips (such as floating guidance and step-by-step pop-ups). These measures effectively lowered the usage threshold and improved students' willingness to continue using the platform.

Teachers' Obstacles in Data Interpretation Ability

Although the platform provides teachers with a range of visual data dashboards (such as class knowledge connection diagrams, individual learning path heat maps, and forum interaction networks), in the early stage, many teachers reported difficulty in quickly extracting instructionally meaningful information and faced a dilemma of seeing the data but not understanding it or experiencing information overload. Some teachers also worried about misinterpreting the data and making inappropriate instructional interventions. To address this issue, the research team conducted several rounds of teacher training, explaining how to link data indicators to teaching objectives using real classroom cases, such as judging students' understanding of a skill through video completion rates or identifying potential learning communities based on forum posting frequency. In addition, a teacher mutual aid community was established to encourage experience sharing and collaborative reflection, gradually improving their educational data literacy.

Information Cocoon Risk Brought by Personalized Recommendation

Although personalized content recommendations based on students' historical behavior on the platform improve learning efficiency, they also introduce the risk of an information cocoon. Some students may engage only with resources closely aligned with their current interests or abilities for extended periods and gradually avoid more challenging or diverse musical content, which limits the expansion of artistic perspective and the development of creativity. To break this closed cycle, the teaching team embedded exploratory tasks into the curriculum design, such as requiring students to listen to and analyze works from non-recommended genres or participate in cross-style group creation projects. Teachers also regularly launched open topics in the platform forum to guide students toward underrepresented but high-value resources on the platform.

Privacy Concerns and Trust Issues of Data Tracking

Continuous recording of learning behaviors (such as click trajectories, duration of stay, and frequency of interaction) provides a foundation for personalized services but also raises concerns among some students about privacy and monitoring. As a result, some students deliberately reduced their use of the platform or engaged less authentically in interactions, which weakened data validity and the accuracy of instructional feedback. To build trust, teachers transparently explained the purpose, scope, and boundaries of data collection at the beginning of the course and clearly emphasized that all behavioral data were used solely to optimize learning support and not as a basis for performance evaluation. At the same time, students were provided with personalized learning feedback reports on a regular basis (such as "your investment in the polyphonic practice module is higher than the class average"), transforming abstract data into concrete evidence of growth. This student-centered communication strategy effectively reduced anxiety and enhanced students' sense of identity and security on the platform.

DISCUSSION

With the deepening of digital transformation in education, big data technology is moving from concept to classroom practice, giving rise to new educational models that integrate data intelligence with subject teaching. The teaching method proposed in this study represents a concrete exploration of this trend in art education. It not only reconstructs teacher–student interaction but also introduces systematic requirements for the educational ecosystem that supports its operation. For school administrators, this model signifies a shift from experience-driven to data-driven management. Traditionally, course quality evaluation depended largely on final grades or teachers' subjective feedback, whereas the new method provides real-time, fine-grained monitoring tools through learning behavior data (such as resource utilization rates, task completion, and interaction depth) continuously generated by the platform. For example, the Academic Affairs Office can identify through the dashboard that the completion rate of video resources in a music theory course is generally low and then organize teaching and research activities to optimize content design. Moreover, when the Academic Affairs Office identifies common characteristics of high-participation courses, these can be promoted as exemplary teaching cases. This approach requires managers not only to understand the educational significance of data indicators but also to establish institutional conditions that support data literacy training, cross-departmental collaboration, and incentives for teaching innovation.

For front-line teachers, this model is both empowering and challenging. Teachers are no longer solely knowledge transmitters but also designers and facilitators of the learning ecology. They can use the platform to gain insights into each student's interests, skill gaps, and collaboration tendencies, enabling more precise task differentiation, personalized guidance, and stimulation of creative potential. For example, a piano teacher may observe that a student repeatedly watches a jazz improvisation video but shows limited forum interaction and can initiate one-on-one online communication to encourage the student to share difficulties. Teachers can also form collaborative groups of students with complementary strengths in rhythm training based on system prompts. However, this model requires teachers to possess basic data interpretation skills, the willingness to adapt teaching strategies flexibly, and openness to technological tools, which underscores the need for continuous professional development support from schools.

For IT managers, this teaching method introduces new responsibilities for infrastructure, operations, and maintenance. The platform must stably support high concurrent access, ensure data security and compliance, and integrate seamlessly with the school's existing educational administration and identity authentication systems. For example, during peak course selection periods at the beginning of the semester, the IT team needs to expand server resources in advance to prevent system congestion. In terms of data storage, the team must ensure that student behavior logs comply with the requirements of the Personal Information Protection Law while implementing anonymization and access control. In addition, IT departments need to establish rapid response mechanisms to resolve technical issues encountered by teachers and students in a timely manner, ensuring that the teaching process is not disrupted. This shift reflects a transition in the IT role from back-office support to an educational technology collaboration partner.

For educational software developers, this study highlights core principles of application development. Technology should serve educational logic rather than dominate the educational process. Developers should not only focus on algorithm accuracy or interface design but also develop a deep understanding of the specific characteristics of music education. For example, the cultivation of unstructured abilities such as intonation, rhythm, and emotional expression is difficult to fully quantify. Therefore, when designing recommendation engines, interfaces for teacher intervention should be incorporated. When constructing knowledge maps, cultural dimensions such as musical styles and cultural context should be included. In data visualization, developers should avoid oversimplifying complex learning processes. A practical example is that, based on teacher feedback, the development team adjusted recommendation logic that had relied solely on click-through rates by incorporating the weight of artistic diversity, ensuring that students are not systematically excluded from classical or traditional music resources due to a preference for popular styles. In summary, the success of this teaching method depends on deep collaboration among managers, teachers, IT personnel, and developers.

CONCLUSION

Against the backdrop of rapid advances in information technology and the digital transformation of education, music education faces dual challenges arising from increasingly personalized student needs and highly fragmented teaching resources. To address these challenges, this study aims to integrate big data technology with Connectivism theory to construct an innovative "big data + music education" teaching model that is learner-centered, data-driven, and connection-oriented. To achieve this goal, a controlled experiment was designed and implemented. The study adopts a mixed-methods approach. On one hand, multidimensional student learning behavior data were collected through an intelligent teaching platform, and analytical techniques such as K-means clustering, multiple linear regression, and Apriori association rule mining were used to quantitatively examine learning patterns, factors influencing academic performance, and effective learning pathways. On the other hand, the effectiveness and acceptance of the new teaching model were evaluated at the level of subjective experience through questionnaires and in-depth interviews.

The research results indicate that the teaching model achieved significant outcomes. First, students' learning behaviors exhibited clear hierarchical characteristics, including active exploration, stable engagement, and passive coping, which provides a basis for implementing targeted intervention strategies. Second, classroom participation was identified as the most important positive factor influencing academic performance, highlighting the central value of highly interactive learning environments. In addition, significant correlations were observed between specific behavior patterns and high academic achievement, while inefficient learning patterns were effectively identified. Finally, students reported high overall satisfaction with the new teaching model and particularly recognized the value of personalized resources and its instructional effectiveness.

The significance of this study lies in the successful construction and validation of a practical paradigm for the technological application of Connectivism theory. This paradigm not only optimizes the teaching process and improves students' learning experience and outcomes but also promotes the transformation of music education from standardized instruction to personalized construction and collaborative creation. However, this study has limitations. The sample size is small and limited to a specific music course at a single university, with relatively homogeneous student backgrounds. This limitation weakens the robustness of the statistical analysis and restricts the applicability and scalability of the proposed teaching and behavior models across broader educational contexts. It remains uncertain whether this model is equally effective across different regions, educational levels, or disciplinary contexts.

Future research should be expanded across multiple dimensions. First, at the empirical level, broader coverage should be achieved through multi-center controlled experiments across regions and educational stages, with systematic collection of behavioral data from heterogeneous learners to test and refine model adaptability under varying cultural contexts, teaching objectives, and infrastructure conditions, thereby establishing a scalable implementation pathway. Second, at the technical level, platform architecture should be restructured by incorporating modular, lightweight, and flexible designs inspired by distributed microservices and edge computing, enabling deployment tailored to school size, network environment, and teacher capacity while lowering implementation barriers. Finally, at the mechanism level, a multidimensional evaluation system should be developed that encompasses learning effectiveness, teacher workload, system stability, and data ethics, alongside embedding a continuous iterative cycle of design, implementation, feedback, and optimization to ensure that the model maintains its core educational value in large-scale applications while dynamically responding to real teaching needs and ultimately promoting the development of a sustainable and reproducible intelligent music education ecosystem.

COMPETING INTERESTS

The authors of this publication declare there are no competing interests.

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